Utilizing Reinforcement Learning with Path Planning Algorithms to Create More Efficient Paths of Travel

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***Abstract*—In this report, the implementation of a reinforcement learning algorithm in conjunction with multiple basic path planning algorithms is investigated to determine if reinforcement learning can efficiently optimize a simulated TurtleBot in a 2D environment based on the manipulation of several key parameters with the performance measured against path length, smoothness, travel time, velocity, and clearance. The simulations under study were conducted with ROS2 in a Gymnasium environment using open-source Turtlebot code and a Nav2 stack. The reinforcement learning algorithm was designed to utilize MPPI and DWB path planners and as a result, optimize them to perform better than their current offline counterparts.**

I. Introduction

A. Current state of the Art

In the industrial environment of Autonomous Mobil Robots (AMRs), path planners are generally deployed en-mass and are not tailored to the customers or environment needs in which they are deployed. This results in good, but not excellent performance, and while easier to maintain on the backend, can limit customer output on the front end. These general deployments often cast a wide net and aim to operate on the "safe side", whether that be wide, nonadjustable clearance, paths around high traffic areas, or creating lanes that avoid foot traffic altogether. In reality, each customer has different needs, and outside of required safety standards, may have certain workplace standards that could offer better performance if utilized properly to tune the AMR deployment. It is obvious that hand tuning any number of parameters for each individual deployment site is unreasonable and the time spent on it would far outweigh any long-term benefits. Additionally, the deployment time for standard AMRs can be several weeks, meaning there is a potential for downtime when looking at brownfield sites. Implementing a reinforcement learning (RL) algorithm paired with machine vision such as LiDAR, could reduce that deployment time from weeks to days, as all the training is done on the back end and there is no need to manually set path layouts.

B. Project Objectives

To reduce the amount of manual input needed to tune path planner parameters, an RL algorithm is deployed into the simulation for a period of training and the following parameters are adjusted to optimize the Turtlebot travel across the simulated warehouse: Clearance, Accuracy, Efficiency, and Smoothness. These four parameters have been previously proven to be statistically significant modifies for both path planners under study [1] [2]. In this context, a dense reward function is custom-built to help direct the RL algorithm through each episode so that when it is deployed, it can be more lightweight in terms of computational load.

C. Background

While modern RL algorithms are a fairly new technology, AMRs and their predecessors, Automated Guided Vehicles (AGVs), are not and have been in use for over 70 years in various forms. Similarly, path planners such as Model Predictive Path Integral (MPPI) and Dynamic Window Approach-Based (DWB) are 10-15 years old. As a result, there is a wide breadth of research that has been conducted to both verify plausiblity as well as serve as a reasonable set of starting points.

D. Path planning

Elbouhy et. al designed a study that consisted of three different path planners that were tested in environments with static and dynamic obstacles and the results were then evaluated to compare algorithms. The study analyzed DWB, MPPI, and RPP (Improved Pure Pursuit) planners and used the following criteria to determine path quality: time, distance, smoothness, minimum clearance, runs completed, and number of recoveries. This study showed that MPPI was the fastest but least smooth, DWB operated with the least clearance, and RPP took the most time and traveled the longest distance [1]. This demonstrates that different planners have different strengths and weaknesses which ultimately led to the integration of both MPPI and DWB into the RL-based simulation built in this project.

Li et. al studied the Dynamic Window Approach (DWA) algorithm and adjusted evaluation parameters in an effort to optimize the path planning. The evaluation function utilized in this paper looked at the heading, the distance between the trajectory and nearest obstacle, and the speed of the robot, along with weighted modifiers of each. A simple environment was setup and the tuning modifiers were adjusted for optimization [2]. In the context of this project, the interest lies in the tuning equation itself; and Li's tuning equation provided a confident proof-of-concept that path planning algorithms can be tuned and optimized on a limited number of parameters and the results are statistically significant and positive.

E. Reinforcement Learning

Qu et. al. presented a method of training an RL policy offline to accelerate the convergence for the online control of MPPI in an unmanned aerial vehicle (UAV). The method they proposed involved utilizing the Distributional Soft Actor-Critic (DSAC) algorithm to have an RL agent train a pair of neural networks to model distributions for the control costs and rollout guiding policy used directly within the online MPPI control. This improved MPPI by providing better "first guesses" than standard MPPI. The results of their experiments showed that their method converged significantly faster, produced smoother trajectories, and provided higher success rate in reaching the goal than standard MPPI [3]. The guiding principle of this paper translates directly to this project to enhance the strengths of existing local planners instead of replacing them by leveraging the strengths of both RL and traditional path planning algorithms to create a more efficient and adaptable path planning solution for mobile robots in complex environments.

Ting et. al. presented a modified Proximal Policy Optimization (PPO) in conjunction with a deep reinforcement learning (DRL) algorithm in order to reduce the time needed to move an agent through an environment. To reduce the number of episodes required and thus the calculation time for each move, they proposed a distance-based weighted dense reward function. The reward function they proposed utilized the euclidean distance to the goal to provide a higher value reward for better steps [4]. Since this reward function is not strictly linked to PPO parameters, the concept can easily translate into both MPPI and DWB that are utilized for the training in this project.

Debnath et al. propose a modular navigation framework for navigation that combines a classical global planner with a DRL based local planner. The global planner generates waypoints and assigns Gaussian soft costs around detected pedestrians to anticipate their future positions, while the trained local planner dynamically follows these waypoints while avoiding imminent collisions in real time. The reward function used to train the local planner includes eight terms covering goal reaching, collision penalties, waypoint tracking progress, pedestrian proximity avoidance, and a small per-timestep penalty to encourage efficiency. Through experiments, their hybrid system outperformed both a traditional DWA planner and a standalone RL baseline, with the RL-augmented system achieving a notably lower collision rate [9]. This paper is directly relevant to the dynamic parameter tuning approach adopted in this project where the RL agent observes local state and adjusts planner parameters throughout the navigation process rather than making a single configuration decision upfront.

Rana and Kaveendran implemented a PPO-based DRL agent to train a TurtleBot3 navigation policy from scratch within a ROS2 environment. The agent observes the robot's surroundings through a LiDAR scan and directly outputs normalized linear and angular velocity commands at each timestep, making it the sole navigation controller rather than an augmentation of an existing planner. Training was conducted using a two-term reward function including a small positive reward for forward progress and a large negative penalty for collisions. The trained policy resulted in better performance, and behavior improvements such as slowing near tight spaces [10]. This paper is particularly relevant as a validation of a similar implementation being used in this project: ROS2, Gymnasium, PPO, and a TurtleBot platform. The key distinction is that their RL agent replaces the planner entirely by outputting direct velocity commands, whereas this project's agent acts as a meta-controller. Their training time was approximately four hours for a significantly harder problem (the agent must learn full navigation from raw sensor data), providing reassurance that the team's proposed approach is viable.

F. Other Noteworthy Resources

In addition to the papers described in detail above, there were several other papers that were reviewed which provided supplemental information for shaping the project. [5] and [6] provided metrics and insights on planner path evaluation which was utilized as inspiration for the evaluation metrics implemented in this project. [7] and [8] demonstrated more examples which utilized RL within some sort of path planning framework for a mobile robot (including both direct and indirect RL control), again reinforcing the possibility of the concept explored in this project.

While there are several papers which demonstrate the use of RL in path planning, no papers were identified which explicitly utilized RL to configure the parameters for a given local path planning algorithm. This is a critical gap in the literature and study of this field, as it is well established that the performance of local path planners can be highly sensitive to their parameter settings, and that optimal settings can vary significantly across different environments and use cases. Meanwhile, some of the examples utilizing direct RL control are not very extensible as these methods tend to only train the RL agent to operate in a specific environment; and re-training for more complex environments can be very costly. By leveraging RL to automatically tune these parameters, this project theoretically fills this gap and demonstrates a novel approach to optimizing path planning performance in a way that is adaptable to the specific needs of different deployment sites.

II. Materials and Methods

A. Simulation setup

In order to reduce the scope and complexity of the required programming of this project, an RL agent was implemented within ROS2. Using ROS2 prevents needing to develop a simulation environment and navigation stack from scratch; which allows for more dedicated effort to developing the RL infrastructure and evaluating performance. Additionally, the simulation environment is constrained to a 2D grid with a TurtleBot robot model in order to simplify the state space and holonomic constraints for the path planners and RL infrastructure, again to avoid adding unnecessary complexity.

The environment was designed to be cross-platform and reproducible, leveraging Docker to encapsulate all dependencies and ensure consistent behavior across the team's Mac, Windows, and Linux development machines. The environment is structured as a two-layer Docker image: a base capstone_ros_layer image that installs all system-level dependencies (ROS2, Nav2, TurtleBot4 description packages, and visualization tools), and a top-level rbe_capstone image that layers in the project's ROS2 workspace. Separating dependency installation from application code allows the more expensive base layer to be rebuilt infrequently while code iteration remains fast.

The simulation runs on ROS2 Jazzy with the CycloneDDS middleware layer for inter-node communication. The robot model used is the TurtleBot4, whose URDF is loaded via `xacro` and published to the `/robot_description` topic by the `robot_state_publisher` node. Navigation is performed in a 2D occupancy grid environment using the Nav2 stack, which includes `nav2_map_server` for loading prebuilt maps and `nav2_lifecycle_manager` for managing the lifecycle of navigation nodes. Maps are stored as paired `.pgm` and `.yaml` files and are selectable at launch time, enabling the team to quickly swap between different warehouse environments as the project progresses.

Visualization is handled by RViz2, which renders the robot model, occupancy map, and planned paths. For the purpose of this project, RViz2 is chosen over Gazebo for visualization because it is more lightweight and the scope of this project does not require a full physics simulation. Gazebo will be consistently slower in launching and running episodes, which is not ideal for iterative tests and tuning of the RL algorithm. RViz2 provides sufficient visualization capabilities for the 2D occupancy grid and planned paths while allowing for faster iteration cycles. To further speed up iterative development, launch files are configured to bring up all required nodes—the robot state publisher, static TF broadcasters, map server, lifecycle manager, and RViz2—making it straightforward to start up the full environment with a single command.

B. RL Architecture

The RL component of the project is a trainable RL agent that tunes planner parameters directly through the Nav2 stack. Independent from the specific path planning algorithm, two design options were initially explored for RL architecture of the states: a one-time parameter tune, or a dynamic parameter tuning. It was decided that the implementation and expected results of a dynamic RL application would be favorable and feasible within the scope of this project. With this approach, the RL agent would be responsible for dynamically tuning the local planner's parameters throughout the navigation process where the state (observation) vector includes the robot's current pose, velocity, distance from the goal, and the obstacle cost of the surrounding environment. The actions continuously adjust the parameters (via a continuous action space) at each timestep. To support frequent observations and actions throughout the navigation process, the reward function is designed to encourage efficient, smooth, and safe navigation by providing positive rewards for progress towards the goal and negative rewards for collisions, long travel times, and time spent in close proximity to obstacles. The goal of these rewards is to encourage the agent to learn how to guide the robot through crowded environments as smoothly and quickly as possible.

1) Training Architecture: Training is driven by the Stable-Baselines3 (SB3) library, which interfaces with the simulation through a custom Gymnasium environment (set up in env.py). SB3 is widely used for training RL agents and provides a variety of algorithms and utilities for training and evaluation, but one challenge that we had to consider is that it is designed for synchronous environments. This project heavily relies on ROS2, which is fundamentally an asynchronous setup with multiple nodes communicating through topics and services. The main training loop needs to block and wait for new observations and rewards from the ROS2 environment, but also needs to be able to control the asynchronous ROS2 nodes and services to set up and run each episode. To bridge this gap, a state machine-like architecture was implemented to allow the main training loop to operate synchronously with each episode. This was achieved by setting up a thread queue to put the main training loop in charge of the episode flow, where the loop listens to the metrics from a ROS2 topic, and then issues service calls to control the episode flow (start episode, switch planners, update parameters) based on those metrics.

A per-step exponential moving average (EMA) with $\alpha = 0.3$ is set up to smooth the raw policy output before it is sent to the navigation stack. The smoothed action vector is appended to the observation so the policy can reason about what parameters are currently active. This is important because the reward function needs to encourage smooth parameter changes, and the policy needs to learn how drastic parameter changes can lead to negative rewards. The EMA helps to mitigate this issue by smoothing out the policy's output, but it also means that the policy needs to learn how its actions will be smoothed over time.

The actual learning for the policy was accomplished using SB3's Soft Actor-Critic (SAC) agent. While most of the reviewed literature used the PPO algorithm, we found that PPO is not suited for converging quickly enough within the environment and time constraints of this project. While PPO

may not be a bad option overall, speeding up PPO learning for this particular state and action space would require extensive hyperparameter tuning — especially with the parameters related to exploration since the action space is so diverse. And even with extensive hyperparameter tuning, the exploration strategy and on-policy nature of PPO may not be ideal for significantly improving learning within this project scope. On the other hand, SAC is an off-policy algorithm with a much stronger emphasis on exploration [11]. This makes the algorithm incredibly sample efficient, which is helpful for environments like ours that are "expensive" to sample (e.g. feedback for each action within our environment is built on multiple complicated layers of asyncronous software). SAC ensures that a wide variety of planner parameter configurations are continually explored and evaluated without degrading the quality of the learned policy.

Checkpoints are saved every 2000 environment steps using SB3's `CheckpointCallback`. The SAC replay buffer is saved alongside the policy weights so that training can be resumed without discarding previously collected transitions. If a checkpoint is resumed but its replay buffer file is missing, the `learning_starts` threshold is extended by an additional 500 steps to prevent gradient updates on an empty buffer.

2) Reward Function: The reward function is built around five per-step terms (dense rewards) and two terminal terms (sparse rewards). Each term targets one of four relevant navigation quality criteria used for evaluation throughout the literature: efficiency, smoothness, accuracy, and clearance.

The primary signal is a progress reward proportional to the reduction in distance to the goal each timestep, which gives the agent a continuous incentive to move forward rather than waiting for a sparse goal signal at the end of the episode. Three penalty terms counterbalance this: a small penalty for deviating laterally from the global plan encourages the agent to stay on the intended path; a penalty on angular velocity discourages unnecessary spinning; and a graduated proximity penalty activates when the planned path ahead of the robot enters an inflated obstacle region on the costmap, increasing as the path gets closer to an obstacle. Unlike a binary collision flag, the graduated penalty provides a smooth signal that the agent can learn from without being destabilized by sudden large negative rewards. A flat time penalty is also applied every timestep regardless of other conditions, making overall navigation speed the dominant optimization pressure.

At the end of each episode, a goal bonus of $+100$ is awarded if the robot successfully reaches its destination, and a failure penalty of -50 is applied if the episode ends without reaching the goal. The larger bonus relative to the penalty is intentional—it keeps the agent willing to attempt difficult routes rather than playing it safe and timing out.

3) Training and Evaluation Tools: To support debugging during both training and evaluation, a live parameter visualization tool was developed. It subscribes to a topic containing the Nav2 parameters being tuned and renders each tunable parameter as a horizontal bar dial in a real-time matplotlib window, updating at 10 Hz. Each dial shows the full valid range for that parameter, a tick mark at the preset baseline value, and a colored fill indicating the current value — blue if the agent has pushed the parameter above baseline and orange if below. An example of the visualization is shown in Figure 1.

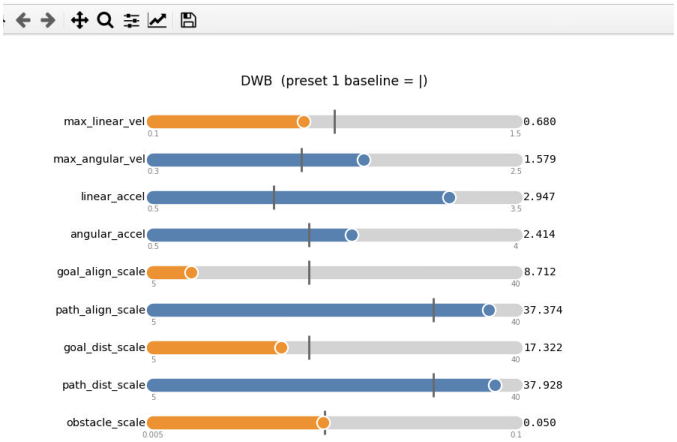


Fig. 1: Live parameter dial visualization during a training run. Each bar shows the current value relative to the preset baseline (vertical tick) and the full tunable range. Blue indicates the agent has pushed a parameter above baseline; orange indicates below.

The tool proved useful for catching two failure modes that were not obvious from reward curves alone. In one case, the agent collapsed all parameters toward one extreme of their range and kept them there, which the dials made immediately apparent. In another case, the parameters were not changing at all between steps, which pointed to a bug in how actions were being applied rather than a training issue. Having a direct view of what the agent was actually doing to the planner at each step made these problems straightforward to diagnose.

C. Performance Evaluation

In order to assess the impacts of incorporating RL within the local path planning process, a set of path quality metrics was used to evaluate the performance of the RL-augmented planner against a baseline planner with standard parameters across a variety of scenarios (e.g. different combinations of maps, start poses and goal poses). The path quality evaluation metrics will be based on the following criteria:

- Clearance from obstacles
- Accuracy (success rate in reaching the goal, amount of deviation from global path plan, etc.)
- Efficiency (total time and distance to goal)
- Smoothness (heading angle variance, velocity variance, etc.)

Since each of the criteria listed above was incorporated directly into the reward function for training the RL agent, the same reward function was used to evaluate the overall "score" of both the RL-augmented and baseline planners. Since the reward is already computed for each step through the

custom gymnasium environment, the performance evaluation was accomplished by having both the RL-augmented planner (operating with a frozen policy) and the baseline planner step through the custom gymnasium enviornment over the same predefined set of episodes.

D. Code Architecture

——PRESENT CODE FLOW / BREAKDOWN OF KEY FILES——

1) Planner control and Episode Runner: These two ROS2 nodes (planner_controller.py and episode_runner.py) work in concert with each other and act as the backbone for the episodic training that is needed for the RL agent as well as building a basic baseline to compare it against. The planner node is able to dynamically switch between the DWB and MPPI planners and then loads and maps the correct parameters (linear and angular acceleration and velocity, as well as required distances) to pass into the episode runner. It also acts as a go-between to stored and communicate parameters as they change throughout the RL training and pushes them out to the episode runner. —-DIAGRAM—-

The episode_runner node, as the name implise, runs a loop of episodes for both the untrained (benchmark) and trained (RL mode) path planners. It sets up the simulation environment including the map and Turtlebot, and metrics for data collection. During the episodes and once the Nav2 finishes, it fetches the designed performance metrics for analysis and comparison. After each timestep in the simulation, this node works with the planner to capture the required data for the rest of the RL architecture. —-DIAGRAM—-

III. Experimentation

A. Standard Path Planners

To generate an accurate baseline and determine the effectiveness of the RL agent, —-XX—- simulations were run with the off-the-shelf MPPI planner and DWB planner. Each simulation was evaluated for clearance, accuracy, efficiency, and smoothness and the results for both can be seen below. The MPPI had an average ——Placeholder for raw data —— ——TABLE / GRAPHICS ———.

B. RL-Trained Planners

For the RL agent, it was first trained —-XX—- and then deployed in the same environments as the above planners. The path was then evaluated against the same four qualifies and comparisons were drawn. ——Placeholder for data —— —— TABLE / GRAPHICS ———. A detailed analysis of the data can be seen below in the Results section.

IV. Results

After conduction all ——NUMBER—- of experiments with both standard and RL-driven planners, it was made clear that ——BIG CONCLUSIVE RESULT——

——GRAPHIC OVERLAYS OF STANDARD / RL PERFORMANCE FOR CLEARANCE, ACCURACY, EFFICIENCY, AND SMOOTHNESS——

Figures 2 and 3 below showcase the current state of the simulation environment as described above.

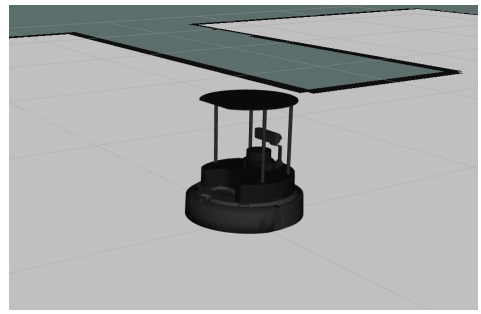
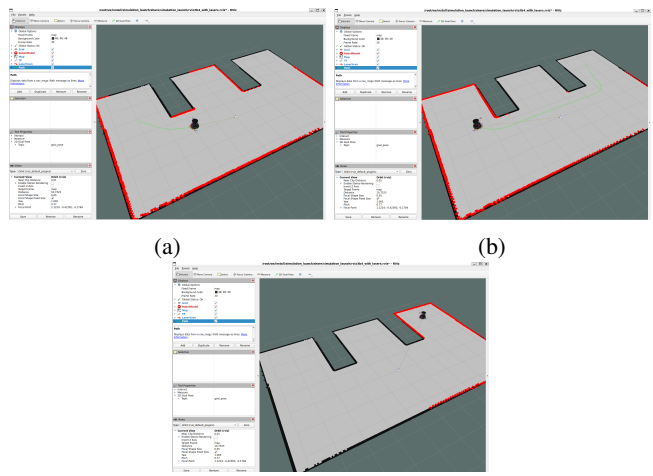


Fig. 2: TurtleBot4 model loaded in RViz2 with a simple occupancy grid map environment for demonstration purposes.



(a) (b) (c)

Fig. 3: TurtleBot4 model scanning and moving through the environment with goal poses set directly through RViz2. In 3a, the robot moves along the generated path (in green) from its initial pose to the first goal pose (in the upper left corner). Then in 3b, a new goal pose was commanded, resulting in a new path (to the upper right corner). Finally, in 3c, the robot reaches the final goal pose. All this navigation was done through manual control; and this sort of process will need to automated in order for the RL agent to be trained. Additionally, as the robot travels from 3a to 3c, it's clear from the LiDAR model scans (in red) that different parts of the environment become occluded. This is an important aspect of the environment that the RL agent will need to learn to account for when tuning the planner parameters.

V. Discussion

A. Implications of Results

B. Future Work

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APPENDIX

A. Source Code

All the code for this project can be found in a Git repository here: [https://github.com/SushiRaj99/rbe-capstone] or seen below.